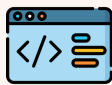


Context Window Constraint



Limited
Working
Memory



No Role Specialization



Manager



Planner



Coder

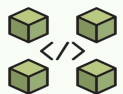


Reviewer



Tester

Single Agent Limitations



- Compile and Test
- Runtime and Profiling
- Simulation / Fuzzing



Repositories and
Databases

Execution, Feedback and Synchronization



File-only



Execution-based



Repository-based

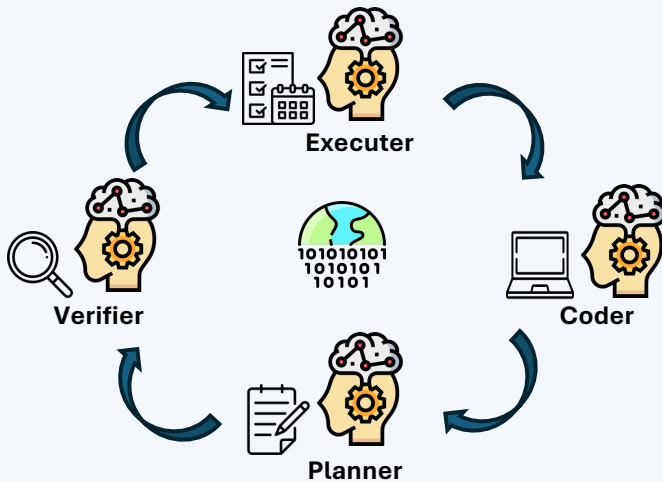


Blackboard

Shared Code-Centric Harness Substrate

Program-State Convergence

- Correctness
- Security
- Performance
- Consensus



Functional Role Specialization

Interaction Modes



Debate



Critique



Adversarial



Collaborative

Heuristic Topologies



Chain



Cyclic



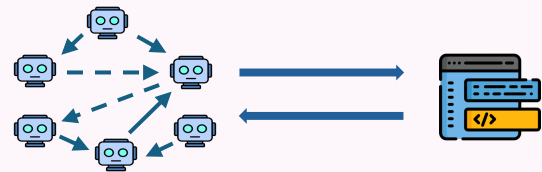
Hierarchical



Star

Adaptive Object-driven Topologies

- Dynamic Agent Pool Scaling
- Feedback-driven DAG restructuring
- Runtime Self-reorganization



Multi-Agent Collaboration